Zhe Liu

Postdoctoral Researcher, Department of Computer Science,
The University of Hong Kong
Email: happinesslzcq@gmail.com URL: Google Scholar HomePage: MyPage

(i) Academic qualifications:

- 2025-Now, Postdoctoral researcher at the Department of Computer Science of the University of Hong Kong under the supervision of Prof. Hengshuang Zhao
- 2020-2025, PhD in the School of Electronic Information and Communications at Huazhong University of Science and Technology under the supervision of Prof. Xiang Bai

(ii) Previous academic positions held:

- Jan. 2022 – Dec. 2024 Intern at Baidu Vision Technology Department and research on multi-modal 3D object perception in autonomous driving
- Sep. 2020 - Dec. 2020 Intern at Huawei Noah's Ark and research on the design of end-to-end 3D object tracking algorithms

(iii) Present academic position held:

My research focuses on embodied AI and autonomous driving, including 3D perception, vision-language-action models, and world models. I have published **40+ papers** in leading venues, including NeurIPS, ICML, CVPR, ECCV, and TPAMI. Google Scholar reports **2,600 citations** and an h-index of 18.

My current research aims to build general-purpose embodied intelligence through:

1. **Embodied Data Engine**, including human data collection, human-to-robot transfer, and corner-case feedback;
2. **Vision-Language-Action Models, World-Action Models, and VLN Systems** for manipulation, navigation, and long-horizon tasks;
3. **4D World Models and Embodied Simulation** for policy learning, virtual evaluation, and sim-to-real transfer;
4. **Efficient and Streaming Embodied Inference**, including distillation, acceleration, and real-time deployment;
5. **Intelligent Robotic OS and Cross-Embodiment Deployment**, unifying interfaces and foundation models.

(iv) Selected publications: *(*: Co-first author; ✉: Corresponding author)*

Section A: Representative 3D perception research

1. **Zhe Liu**, Tengteng Huang, Bingling Li, et al. EPNet++: Cascade Bi-directional Fusion for Multi-Modal 3D Object Detection. **TPAMI 2022.**
2. **Zhe Liu**, Xin Zhao, Tengteng Huang, et al. TANet: Robust 3D Object Detection from Point Clouds with Triple Attention. **AAAI 2020. (Oral)**
3. Tengteng Huang*, **Zhe Liu***, Xiwu Chen, Xiang Bai. EPNet: Enhancing Point Features with Image Semantics for 3D Object Detection. **ECCV 2020.**
4. **Zhe Liu***, Jinghua Hou*, Xingyu Wang, et al. LION: Linear Group RNN for 3D Object Detection in Point Clouds. **NeurIPS 2024.**

Section B: Recent embodied AI and world models (†: Project lead)

1. **Zhe Liu***, Jinghua Hou*, Yuxiang Lu, et al. StreamPI: Streaming Multimodal Temporal Modeling for Vision-Language-Action Models. **NeurIPS 2026.**
2. Yuxiang Lu, **Zhe Liu**†, Xianzhe Fan, et al. FASTER: Rethinking Real-Time Flow VLAs. **NeurIPS 2026.**
3. Xirui Li, **Zhe Liu**†, Xiaoqing Ye✉, et al. FlowR2A: Learning Reward-to-Action Distribution for Multimodal Driving Planning. **NeurIPS 2026.**

4. Junwei Luo, Shuai Yuan, Zhenya Yang, Yansheng Li, **Zhe Liu**, Hengshuang Zhao. EO-WM: A Physically Informed World Model for Probabilistic Earth Observation Forecasting. **NeurIPS 2026.**
5. Runhui Huang, Qihui Zhang, **Zhe Liu**, et al. Read It Back: Pretrained MLLMs Are Zero-Shot Reward Models for Text-to-Image Generation. **NeurIPS 2026. (Spotlight)**
6. Ruihua Han, Rui Gao, **Zhe Liu**, et al. SRL-MPC: Shape-Aware Reinforcement Learned Model Predictive Control. **NeurIPS 2026.**
7. **Zhe Liu**, Runhui Huang, Rui Yang, et al. DrivePI: Spatial-aware 4D MLLM for Unified Autonomous Driving Understanding, Perception, Prediction and Planning. **CVPR 2026.**
8. Zhenya Yang, **Zhe Liu**[†], Yuxiang Lu, et al. GenieDrive: Towards Physics-Aware Driving World Model with 4D Occupancy Guided Video Generation. **CVPR 2026.**
9. Runhui Huang, Jie Wu, Rui Yang, **Zhe Liu**, Hengshuang Zhao. AlphaGRPO: Unlocking Self-Reflective Multimodal Generation in UMMs via Decompositional Verifiable Reward. **ICML 2026.**
10. Shijun Shi[*], **Zhe Liu**[*], Zhi Hou, et al. Atomic-WAM: Execution-Grounded Atomic-Motion Reasoning for World Action Models. **arXiv:2610.20157, 2026.**
11. Jinghua Hou, **Zhe Liu**[†], Hengshuang Zhao[✉]. DVD: Dynamic Vector Decoding for Efficient MLLM-based Perception. **arXiv:2610.12266, 2026.**

(v) Professional qualifications, prizes and awards:

- Fall 2019: **Master National Scholarship Winner**, School of Artificial Intelligence and Automation, HUST.
- Spring 2020: **"NavInfo" Scholarship Winner**. **(Top 3 of 300+).**

(vi) Research output:

- Three-dimensional object detection method and system based on weighted channel features of point cloud. *Zhao Xin, Huang Kaiqi, **Liu Zhe.*** **(U.S. Patent)**
- A robust three-dimensional object detection method based on ternary attention mechanism. *Bai Xiang, **Liu Zhe**, Zhou Yu, Huang Tengteng.* **(Chinese Patent,Authorized**)
- A three-dimensional object detection method, device and system based on DETR structure. *Bai Xiang, **Liu Zhe**, Hou Jinghua.* **(Chinese Patent,Submitted)**
- A three-dimensional object detection method, device and system. *Bai Xiang, **Liu Zhe**, Hou Jinghua, Wang Xinyu, Zhao Hengshuang.* **(Chinese Patent, Submitted)**

(vii) Leading Projects (2025-2026):

1. **AI Glasses:** Efficient 3D perception, spatial understanding, and memory retrieval (Huawei).
2. **4D MLLM:** Unified vision-action and vision-language-action modeling with a 0.5B model (Huawei).
3. **4D World Model:** Occupancy-guided, physically aware video generation (Huawei).
4. **World Model + RL:** Efficient world modeling for autonomous driving (Changan).
5. **Streaming VLA:** Memory-dependent, spatially aware policy inference (ACE Robotics).
6. **Ongoing:** Efficient world models, efficient VLA, and embodied OS.

(viii) Academic services:

- ***Area Chair:*** ICLR 2027 (invited), ICLR 2026.
- ***Conference Reviewer:*** NeurIPS, ICML, ICLR, CVPR, ICCV, ECCV, AAAI, ICRA.
- ***Journal Reviewer:*** TPAMI, TIP, TITS, TCSVT, RA-L.